SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA0307	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	CSE-AI	**Date:**	

Code of Conduct

I **A.Sai Rohit (192472144)** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **A.Sai Rohit**

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

1. A large-scale search engine indexing system processes user queries containing word variants such as "analyzing", "analysis", and "analytical". The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.

2. An AI-based sentiment analysis platform receives product reviews containing words like "disagree", "agreement", and "agreeable". The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.

3. A text analytics pipeline in a news aggregation system handles documents containing terms such as "govern", "government", and "governance". The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown.

4. A document classification system processes input words such as "activate", "activation", and "reactivation" to improve semantic grouping and indexing.
Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.
Analyze how prefix addition and suffix transformation affect meaning and word class.
Apply the process to decompose each word and map them to a unified base representation.
Derive the morphological structure and normalized root for each input word.

5. A search optimization module processes input words such as "create", "creates", and "creating" to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.
Identify suffix patterns and map all variations to a common base form without altering meaning.
Apply the process to extract root forms and classify each transformation.
Determine the morphological breakdown and normalized output for each word.

Scenario 1: Search Engine Indexing – analyzing / analysis / analytical

1.Understanding of the Scenario

A modern search engine processes millions of user queries every day. Different users often search for the same concept using different grammatical forms of a word. For example, one user may search for "analyzing data", another for "data analysis", and another for "analytical methods". Although these words share the same underlying concept, they differ in their morphological structure and grammatical category.

If the search engine treats each form as an entirely separate word, relevant documents may not be retrieved, reducing search accuracy and user satisfaction. Therefore, the indexing system must perform morphological analysis to identify the common root shared by these words and normalize them into a unified representation. At the same time, it should preserve useful linguistic information such as part of speech (verb, noun, adjective) and grammatical features for ranking and contextual understanding.

The objective is to design a morphological processing approach that:

- Decomposes each word into its root and affixes.
- Identifies whether the affixes are inflectional or derivational.
- Groups semantically related forms under a common indexing key.
- Retains grammatical information as metadata to improve search precision and ranking.

This approach enables the search engine to return relevant documents regardless of which grammatical form the user enters.

2.Identification of Key Issues

Several linguistic and indexing challenges arise when processing related word forms such as analyzing, analysis, and analytical.

The first challenge is morphological variation. The same concept appears in multiple surface forms due to grammatical and word-formation processes. Without normalization, the search engine would create separate index entries for each form, resulting in fragmented search results.

The second challenge is the difference in word classes. The three words belong to different parts of speech: analyzing is a verb, analysis is a noun, and analytical is an adjective. Although they represent the same concept, each performs a different grammatical function. The indexing system must therefore recognize both their differences and their semantic relationship.

Another important issue is distinguishing between inflectional and derivational morphology. Inflectional morphology changes only the grammatical form of a word without altering its core meaning, whereas derivational morphology creates new words or changes the word class. Correctly identifying these differences is essential for effective indexing.

The system must also create a unified normalized representation so that all related word forms point to the same conceptual index while preserving grammatical information for advanced ranking and language processing. Proper normalization improves retrieval efficiency, reduces index size, increases recall, and ensures that users receive relevant search results regardless of which grammatical variant they use.

3.Application of Morphological Concepts

Morphological processing involves analyzing the internal structure of words by separating them into roots and affixes. This helps identify the relationship between different word forms and enables efficient normalization during indexing.

The common lexical root across all three words is **analy**-, represented by the base verb **analyze**. This root originates from the Greek language and refers to examining or breaking something into smaller parts for better understanding.

The word **analyzing** is formed by attaching the suffix -**ing** to the base verb **analyze**. The suffix -**ing** is an inflectional suffix because it indicates the present participle or progressive aspect without changing the word class or core meaning. The word therefore remains a verb.

The word **analysis** is a noun derived from the same root. It is formed using the derivational suffix -**sis**, which converts the action expressed by the verb into the concept or result of that action. Since the suffix changes the grammatical category from a verb-related concept to a noun, it is classified as derivational morphology.

The word **analytical** is an adjective formed using the derivational suffixes -**tic** and -**al**. These suffixes transform the noun concept into an adjective that describes something related to analysis. Because the word class changes from noun to adjective, these suffixes are also classified as derivational.

During indexing, the morphological analyzer identifies these relationships and maps all three words to the same conceptual root while storing their grammatical categories as metadata. This allows the search engine to retrieve relevant documents for all related word forms while preserving useful linguistic information.

4.Morphological Decomposition (Decision Output)

Word	Root/Base	Affix(es)	Type	Word Class
analyzing	analyze	-ing (suffix)	Inflectional	Verb (progressive)
analysis	analy-	-sis (suffix)	Derivational	Noun
analytical	analy-	-tic + -al (suffixes)	Derivational	Adjective

5.Decision Making / Rationale

Based on the morphological analysis, the search engine should adopt a normalization strategy that balances retrieval accuracy with linguistic precision.

All three words—analyzing, analysis, and analytical—should be linked to a common conceptual root such as **analyze** or **analysis**. This ensures that documents containing any of these forms can be retrieved when a user searches using any one of them.

The inflectional suffix -**ing** should not generate a separate index entry because it only indicates grammatical tense or aspect. Instead, the system should remove this suffix during indexing and store the grammatical feature as metadata if required.

Derivational forms should not be discarded because they represent different parts of speech that provide valuable contextual information. Instead, they should remain connected within the same semantic cluster while preserving their individual grammatical categories. This enables more accurate ranking based on user intent.

For example, a query containing **analysis techniques** may prioritize noun-based documents, while **analytical methods** may rank adjective-based content more highly, and **analyzing data** may retrieve procedural or action-oriented documents. Maintaining part-of-speech information therefore improves semantic search, query expansion, document classification, and overall retrieval performance.

This normalization strategy reduces duplicate index entries, improves search recall, preserves contextual meaning, and provides users with more relevant search results.

6.Conclusion (Normalized Representation)

Morphological processing is an essential component of modern search engine indexing because it identifies relationships among different grammatical forms of the same concept. By extracting roots, identifying affixes, and distinguishing between inflectional and derivational morphology, the indexing system can normalize related words into a common semantic representation.

For the given inputs, **analyzing** is identified as an inflectional verb form, **analysis** as a derivational noun form, and **analytical** as a derivational adjective form. Although these words differ in grammatical structure, they all represent the same underlying concept of analysis.

Therefore, all three words are grouped under the normalized indexing key **"analyze / analysis."** The system also preserves metadata describing each word's grammatical category and morphological characteristics. This information improves search ranking, contextual understanding, and semantic retrieval while reducing index fragmentation and storage redundancy.

Scenario 2: Sentiment Analysis Platform – disagree / agreement / agreeable

1.Understanding of the Scenario

An AI-based sentiment analysis platform processes thousands of customer reviews to determine whether opinions expressed about products or services are positive, negative, or neutral. Users often express similar ideas using different morphological forms of a word. For example, words such as disagree, agreement, and agreeable are derived from the same base concept but differ in meaning, grammatical role, and sentiment orientation because of the addition of prefixes and suffixes.

If these variations are treated as unrelated words, the sentiment analysis system may incorrectly classify opinions. Therefore, the system must perform morphological parsing to identify the base form, detect prefixes and suffixes, distinguish derivational transformations, and understand how each affix changes meaning. This enables consistent semantic interpretation while preserving the sentiment expressed by each word.

2.Identification of Key Issues

The system faces several challenges while processing these words.

- The words share a common base but differ because of derivational prefixes and suffixes.
- The prefix dis- changes the meaning from positive agreement to disagreement or opposition.
- The suffix -ment converts the verb into a noun, while -able converts it into an adjective.
- Different word classes (verb, noun, adjective) must be recognized while maintaining their semantic relationship.
- The system must normalize related words without losing sentiment information introduced by derivational changes.
- Accurate prefix detection is essential because prefixes may reverse or significantly modify the original meaning.

3.Application of Morphological Concepts

Morphological analysis separates each word into prefixes, roots, and suffixes to understand its structure and meaning.

The common lexical base is agree.

The word disagree consists of the negative prefix dis- attached to the base verb agree. The prefix changes the meaning to express opposition while the word remains a verb. Since the meaning changes, this is a derivational process.

The word agreement is formed by attaching the noun-forming suffix -ment to the base verb agree. The suffix changes the grammatical category from a verb to a noun and represents the state or result of agreeing.

The word agreeable is formed by attaching the adjective-forming suffix -able to the base verb agree. This derivational suffix changes the word into an adjective describing someone or something pleasant or willing to agree.

The morphological parser identifies these transformations, extracts the common base, and stores both semantic and grammatical information for sentiment analysis.

4.Morphological Decomposition (Decision Output)

Word	Prefix	Root	Suffix	Type	Sentiment Effect
disagree	dis- (negation)	agree	—	Derivational	Reverses polarity (negative)
agreement	—	agree	-ment (noun)	Derivational	Neutral/Positive (noun form)
agreeable	—	agree	-able (adj.)	Derivational	Positive (adjective form)

5.Decision Making / Rationale

The sentiment analysis system should normalize all three words under the common base agree while preserving information about derivational changes.

The prefix dis- should be stored as semantic metadata because it reverses the polarity of the base meaning and directly influences sentiment classification. The suffixes -ment and -able should also be retained as metadata because they indicate different grammatical categories and contextual meanings.

By preserving both prefix and suffix information, the system can distinguish between positive, neutral, and negative expressions while maintaining semantic consistency. This improves sentiment prediction, feature extraction, opinion mining, and text classification.

6.Conclusion (Normalized Representation)

Morphological analysis enables the sentiment analysis system to identify the shared base agree while recognizing how derivational prefixes and suffixes modify meaning and grammatical category.The words disagree, agreement, and agreeable are all linked to the normalized base agree, while metadata records the presence of the negative prefix and the noun or adjective suffixes. This approach improves semantic interpretation, preserves sentiment polarity, and increases the overall accuracy of opinion classification.

Scenario 3: News Aggregation Topic Modelling – govern / government / governance

1.Understanding of the Scenario

A news aggregation system processes articles from multiple sources covering politics, administration, economics, and public affairs. Similar concepts often appear in different morphological forms such as govern, government, and governance. Although these words refer to the same general concept of governing, they belong to different grammatical categories and serve different semantic purposes.

Without morphological normalization, the topic modeling system may treat these related words as separate features, reducing clustering accuracy. Therefore, the system must identify the common root, analyze derivational suffixes, and normalize related words under a unified representation while preserving their grammatical distinctions.

2.Identification of Key Issues

The major challenges include:

- Multiple derivational forms represent the same conceptual domain.
- The suffixes -ment and -ance create different noun forms.
- Different grammatical categories must be preserved for semantic understanding.
- The system must avoid duplicate topic representations caused by morphological variation.
- Proper normalization improves document clustering, keyword extraction, and topic modeling.

3.Application of Morphological Concepts

The common base across all three words is govern.

The word govern is the root verb representing the action of ruling or administering.

The word government is formed by adding the derivational suffix -ment, converting the verb into a noun referring to the governing institution. The word governance is formed by adding the derivational suffix -ance, creating a noun that refers to the process or system of governing. Morphological analysis identifies these suffixes, extracts the common base, and maps the words into a shared conceptual cluster while retaining their grammatical roles.

4.Morphological Decomposition (Decision Output)

Word	Root/Base	Affix	Type	Resulting Meaning
govern	govern	— (base verb)	Base form	Action of ruling/directing
government	govern	-ment (suffix)	Derivational	Institution/body that governs
governance	govern	-ance (suffix)	Derivational	Process/system of governing

5.Decision Making / Rationale

The news aggregation system should normalize all three words under the common base govern.The suffixes -ment and -ance should be preserved as derivational metadata because they represent different institutional and procedural meanings. Keeping this information improves topic modeling, semantic clustering, document classification, and keyword ranking.Grouping these forms under a single conceptual node reduces redundancy and strengthens relationships among politically related documents.

Based on the morphological analysis, the news aggregation system should normalize govern, government, and governance under the common base govern. The derivational suffixes -ment and -ance should not create separate conceptual index entries because they represent closely related meanings within the same semantic domain. Instead, these suffixes should be stored as derivational metadata to preserve grammatical information and contextual differences. This normalization strategy improves topic modeling, document clustering, semantic search, and keyword extraction while reducing redundant index entries. It also enables the system to retrieve and group news articles discussing governing institutions, governing processes, and governing actions under the same conceptual topic.

6.Conclusion (Normalized Representation)

Morphological normalization identifies govern as the common lexical base for govern, government, and governance.Although the derivational suffixes create different noun forms, all three words belong to the same semantic concept. Mapping them to the normalized representation govern improves topic clustering, document indexing, semantic search, and information retrieval while preserving grammatical information.

Morphological parsing successfully identifies govern as the common lexical root of govern, government, and governance. Although the derivational suffixes -ment and -ance produce different noun forms with slightly different meanings, they all belong to the same conceptual family related to governing. Therefore, the system maps these words to the normalized representation govern, while preserving derivational information and grammatical categories as metadata. This approach improves topic modeling, document clustering, indexing efficiency, semantic retrieval, and overall accuracy of the news aggregation system by ensuring that related word forms are treated as part of the same semantic concept.

Scenario 4: Document Classification – activate / activation / reactivation

1.Understanding of the Scenario

A news aggregation system collects and processes articles from multiple sources covering topics such as politics, public administration, economics, and governance. Different articles may use related words such as **govern**, **government**, and **governance** to describe the same general concept. Although these words share a common lexical base, they differ in their grammatical categories and morphological structure because of derivational suffixes. If the system indexes these words separately, similar news articles may not be grouped together accurately, reducing the effectiveness of topic modeling and document clustering. Therefore, the system must perform morphological analysis to identify the common root, separate the root from its affixes, recognize derivational layers, and normalize all related words into a unified representation while preserving useful grammatical information.

2.Identification of Key Issues

The news aggregation system encounters several challenges when processing these related word forms. The words **govern**, **government**, and **governance** belong to the same semantic family but have different grammatical roles because of derivational suffixes. The suffix **-ment** creates a noun representing an institution or organization, while the suffix -**ance** forms a noun referring to the process, practice, or system of governing. The system must correctly distinguish these derivational transformations while recognizing that all three words refer to the same underlying concept. Without normalization, documents discussing similar topics may be placed into different clusters, reducing topic modeling accuracy, increasing index redundancy, and affecting semantic search performance.

3.Application of Morphological Concepts

Morphological analysis begins by identifying **activate** as the common lexical base shared by all three words. The word **activate** functions as the base verb representing the action of making something active. The word **activation** is formed by attaching the derivational suffix **-ation**, which converts the verb into a noun representing the process or result of activating. The word **reactivation** contains two derivational layers. The prefix **re**- is added before the base verb to indicate repetition or renewal of the action, while the suffix **-ation** converts the resulting verb into a noun. During parsing, the system separates the prefix, root, and suffix, classifies each morphological transformation, and maps the words to the same conceptual base while preserving grammatical and semantic information.

4.Morphological Decomposition (Decision Output)

Word	Prefix	Root/Base	Suffix	Layer	Type
activate	—	act(ive)	-ate	Layer 1 (base verb)	Derivational (from 'active')
activation	—	activate	-ion	Layer 2 (verb -> noun)	Derivational
reactivation	re- (again)	activation	—	Layer 3 (prefix on noun)	Derivational (layered)

5.Decision Making / Rationale

Based on the morphological analysis, the news aggregation system should normalize **govern**, **government**, and

governance under the common base **govern**. The derivational suffixes -**ment** and -**ance** should not create separate conceptual index entries because they represent closely related meanings within the same semantic domain. Instead, these suffixes should be stored as derivational metadata to preserve grammatical information and contextual differences. This normalization strategy improves topic modeling, document clustering, semantic search, and keyword extraction while reducing redundant index entries. It also enables the system to retrieve and group news articles discussing governing institutions, governing processes, and governing actions under the same conceptual topic.

6.Conclusion (Normalized Representation)

Morphological parsing successfully identifies **activate** as the shared lexical root of **activate**, **activation**, and **reactivation**. Although the addition of the derivational suffix -**ation** changes the word class and the prefix **re-** modifies the meaning by indicating repetition, all three words belong to the same conceptual family. Therefore, they are normalized under the common representation **activate**, while the derivational prefix and suffix information are retained as metadata. This normalization strategy improves semantic grouping, document

Scenario 5: Search Optimization – create / creates / creating

1.Understanding of the Scenario

A search optimization module processes millions of user queries that contain different grammatical forms of the same word, such as **create**, **creates**, and **creating**. Although these words differ in tense and aspect, they all represent the same fundamental action. If the search engine indexes each variation separately, relevant documents may be fragmented across multiple index entries, reducing retrieval accuracy. Therefore, the system must perform morphological normalization by identifying the common base form, removing inflectional suffixes, and preserving grammatical information as metadata. This enables consistent retrieval of documents regardless of which grammatical form users enter in their search queries.

2.Identification of Key Issues

The search optimization system must address several morphological challenges. The suffix **-s** indicates the third-person singular present tense, while the suffix **-ing** represents the present participle or progressive aspect. These suffixes modify only the grammatical form of the word without changing its core meaning or word class. The system must distinguish these inflectional transformations from derivational changes, normalize all word variants to a single base form, and preserve tense and aspect information separately. Proper normalization reduces duplicate index entries, improves retrieval consistency, and enhances search performance.

3.Application of Morphological Concepts

Morphological analysis identifies create as the common base verb shared by all three words. The word create is the root form representing the action itself. The word creates is formed by adding the inflectional suffix -s, which marks the third-person singular present tense while leaving the meaning and grammatical category unchanged. The word creating is formed by adding the inflectional suffix -ing, which expresses the present participle or progressive aspect without changing the word class. During indexing, the morphological parser removes these inflectional suffixes, extracts the base form create, and stores grammatical information such as tense and aspect as metadata.

4.Morphological Decomposition (Decision Output)

Word	Root/Base	Suffix	Type	Grammatical Function
create	create	— (base form)	Base	Infinitive/base verb
creates	create	-s	Inflectional	3rd person singular, present tense
creating	create	-ing	Inflectional	Present participle / progressive aspect

5.Decision Making / Rationale

The search optimization module should normalize create, creates, and creating under the common base create because the suffixes -s and -ing do not alter the core meaning of the word. These inflectional suffixes should not generate separate index entries; instead, they should be retained only as grammatical metadata indicating tense and aspect. By treating all three forms as variations of the same lexical item, the system improves search recall, reduces index redundancy, supports query expansion, and provides more accurate and consistent search results.

6.Conclusion (Normalized Representation)

Morphological normalization identifies create as the shared lexical base for create, creates, and creating. Since the suffixes -s and -ing are purely inflectional, they affect only grammatical features such as tense and aspect without changing the core meaning of the word. Therefore, all three forms are mapped to the normalized representation create, while grammatical information is preserved as metadata. This approach enhances indexing efficiency, improves retrieval accuracy, reduces redundancy in the search index, and ensures that users receive relevant documents regardless of the grammatical variation used in their queries.

Overall Summary Table

The table below consolidates the normalized root/base form and the dominant morphological process identified for each scenario, supporting the clarity and decision-making criteria across the full assignment.

Scenario	Input Words	Normalized Root	Dominant Process
1	analyzing, analysis, analytical	analyze / analysis	Mixed: inflectional + derivational
2	disagree, agreement, agreeable	agree	Derivational (prefix & suffix)
3	govern, government, governance	govern	Derivational (suffix)
4	activate, activation, reactivation	activate	Derivational (layered prefix + suffix)
5	create, creates, creating	create	Inflectional only

Across all five scenarios, distinguishing inflectional morphology (grammatical marking that preserves meaning and word class) from derivational morphology (affixation that changes meaning, polarity, or word class) proved essential for building an accurate, unified normalization layer for indexing, sentiment interpretation, and topic/document classification.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________